Task 4 – BOM

IR Sensor Array (QTR-8 RC)

- We can use a premade ir sensor array for this project instead of manually taking multiple ir sensor modules … basically
- It has the following advantages
    - In the premade ir array the most important advantage is that the 8 pairs come presoldered and with also equal spacing so first thing is we dont need to take the pain to measure the dinstaces and make sure all the 8 modules of ir sensors are fixed and also that there will be vey very less pain that we need to take during the calibration …
    - And also the connections will also be easy since we will get all the pins directly soldered to one singular board so the connections will be more organised and the connections will also not look that shabby
    - Link - https://www.amazon.in/Electron-QTR-8RC-Following-Reflectance-Sensor/dp/B0DYP4JP2V?mcid=8d4085bf35be3cfd955659775a985b49&tag=googleshopdes-21&linkCode=df0&hvadid=709963085510&hvpos&hvnetw=g&hvrand=11556278949578119663&hvpone&hvptwo&hvqmt&hvdev=c&hvdvcmdl&hvlocint&hvlocphy=9183877&hvtargid=pla-2422909312205&psc=1&hvocijid=11556278949578119663-B0DYP4JP2V-&hvexpln=0
    - Price – 240Rs

Microcontroller (ESP 32 WROOM 32 Devkit board)

- We can also use the chip itself .. but using the devkit board is more easy.. So devkit board it is
    - Link - https://www.amazon.in/SP-ELECTRON-ESP32-Development-ESP32-WROOM-32/dp/B0HD7QC26L?crid=2K6Z1AM6WO9VD&dib=eyJ2IjoiMSJ9.H5Qdvll8rdKi4WqCiS9VhdI3iaZbhgtQl5klI5fbAqTaHnH9PVfhgVGmASJWMuZNB3Mxu6roYG4ggagBCZ7XaCWCBcJvx_T-YgyaVu8gdqFZE-4fkuTlUJdxvgzD6RV8qxLBlogARd1CI1B5OTSbrFFtjrvCrieT-9f5GAYHtIQ6vPddJV-NWIbnqPjV12Tw1KI-4X_kkj5czKc-yjeG4fTURmEwLl-IIzfKoPTiqysDBVjEEEs0xjfvRMQ_3V4bB2NMqUA4Y4goWrmo1mJqeYysbLSQQruYxZcDzECG-SM.LJbMbiMqNHSWJKMeUsR8qRuBlAAKLQ8eyhUiBMuj1z0&dib_tag=se&keywords=esp+32+WROOM+32&qid=1792926887&s=industrial&sprefix=esp+32+wroom+3%2Cindustrial%2C385&sr=1-1
    - Price – 500Rs

Motors (2 N20 6v Metal Geared Motors{200rpm} with encoders)

- The main reason of me not using the 300 rpm motor is eventhough it is faster it is harder to tune ..and besides we are not making a line following robot where speed is the priority our priority is accuracy and easier tuning so i think the 200 rpm will do just fine
- And also compered to normal tt plasic geared motors .. the main difference is the gears (which are metal in n20) and also their operating voltages .. and also has an encoder .. to measure the roations of the motor .....
- Basically more control during turnings and intersections .. and far less pain during tuning..
- And the reason for the encoders is because we can map the maze ..
    - Link - https://www.amazon.in/200RPM-Micro-Metal-length-Encoder/dp/B0G49CC6YW?crid=3SMQO1HO4OMWI&dib=eyJ2IjoiMSJ9.IiFODmiFUn4Yi0w6gZSeuVWxcZkYJ94dEZewNoefHkrjb2bYAcOb9CeZA65lPkWxwO9VZsRXGqGKzzTBRmWnNWG_KQbTb4P9ei96Tvu59oFKAkUYg9mikC9RnTjmWncLVR-J90AWtkY4ILPZmjVmC9Qw5FfXPKLCo01mBh7klzYnrcsNNqNrUCVkkp4iEoSLWyaSXoIGLkwZiBwDuQUe47U8_HjukcBQ64aXYlXwMNqihOPqz2uqF6ul9sgEtPwVM0KUNM0Y2o3D3IFUy3bG0rhddLYJk5H5FWoKe0KPNt0.K8yGXiBtykzxvx-28_CDnresWipz3i5TDnyScKML3FM&dib_tag=se&keywords=n20+Geared+motors+200rpm&qid=1792927333&s=industrial&sprefix=n20+geared+motors+200rp%2Cindustrial%2C297&sr=1-1
    - Price 1200Rs
- For Wheels we need some wheels with more traction so something like these below...
    - Link - https://dyoraarobotics.in/product/2
    - Price 269Rs
    - Or we can get soemthing like this also - https://www.theengineerstore.in/products/n20-dc-gear-motor-wheel-42mm?variant=43720257994915&country=IN&currency=INR&utm_medium=product_sync&utm_source=google&utm_content=sag_organic&utm_campaign=sag_organic
    - Price 180Rs

Motor Driver (TB6612FNG Dual DC Motor Driver Module)

- The main reason we are using this is in line folowing bots generally .. we have rapid acceleration and suddeen breaks .. so to have better control we use the above(especially with pid controller)
- And also that the l298n is more bulky
    - Link - https://www.amazon.in/TB6612FNG-Dual-Motor-Driver-PACK/dp/B0GMVYN79G
    - Price – 400 Rs

Miscellaneous Stuff

1. LED Light
    a. For indicating that it finished the maze
    b. LED Toh club se mil hi jaayegi .. khareedne ki kya jaroorat 🥀🥀🥀

c. Link - https://www.amazon.in/Electronicspices-Ultra-Bright-White-Diode/dp/B085BMH3XZ?crid=1RMTZFUN22F45&dib=eyJ2IjoiMSJ9.-cwHRl7D3tkTZrkwgsrs1i33jua-2d8lJZRwI9BjtWLGP9ABlPwH14FOhKKCR6MuO3fQmsvU4sGltYJUJmyPzLZM9jK38sZmActQI_qFlMUIqKdrVHIHMFFXEbPPihfEoqqTIfrY_yinPvGXZcymY0wE8yc5rgmY6cWJVdBJzhoAEuNpfr8qTY4TcGxioZyBPH2rsypDF1YBz72x28HVcDoGMJ9_hbIIx8Yr7cb4CWRpoSdwUbvhTLkVSoaFMrxyupPJmOzO1FUoutb-GAmzl2HX4JfEcN9MQAjus1ui8Qk.dimrmvLh2qJJvHRXkAWKK5Me85e_ld65yIFTocHJjow&dib_tag=se&keywords=single+led+light&qid=1792956319&sprefix=single+led+ligh%2Caps%2C324&sr=8-7

d. Price – 50 Rs

2. A Switch
    a. This too .. club se hi utha lenge
    b. Link - https://www.amazon.in/SP-Illuminated-Electronics-Automotive-Appliance/dp/B0H6F78F97?crid=2I11LVZ0ZXP0V&dib=eyJ2IjoiMSJ9.7VsmIOwBhZtMOOwoROD4Bjy10WjQ5FzPBCHhzmdyUE6cMd8ftbH-x4OLg-ebYPnh6sf8isKFQFg4h1D-A_Dnizx9sy4EIqWXjzP978ocoG8jMRnQITYjNLv6H1vshoRQpmI4nYYaAY3oYaDV5O8MCxwXOIfV6Z6e9krBDPzyyOA_hjQ332DzMAGFyXUlDkkdg08v052mEobbHjn5KZoFAL_9qgDE8AInlkjpZaOfIUyjSaUCRX0Z44boyoUFcTwDDZA0nxcjtwxYmUBcHrF9fXlYpumci1YIr1l-lUO0K0Y.LFP7ZUaqg6tQw8_nfKyGr-sRr4yY7O2Nwdat6xRil6c&dib_tag=se&keywords=small+switch&qid=1792956360&sprefix=smallswithch%2Caps%2C379&sr=8-6

    c. Price – 30 Rs
3. Chasis
    a. We want it to be less bulky ..
    b. Something like below can work (My personal preference btw .. we can pick something else also ..)
    c. Link - https://www.amazon.in/Round-Acrylic-Robot-Chassis-Black/dp/B0F47K9N19?crid=6QN0RTTGZ8YM&dib=eyJ2IjoiMSJ9.856Ifh5W_4uj8KK69ib8xmKDPJgfXLcTyrZHXn6idEqs-x0SGUExU8W2lK3G4QBMDWQF16Tp7IRQs-Prsx9_vSrcQ227eSCYA87jGbkSxE0jejRvb5TOm2J7hjsgJIl_OCVjPiGDDZggFm81PFtcugPmqgUWAKtFfk8wi94JVDsOcCVWUMMnzOK3SnLUv-FnXt6uGSiwnX5Xd9u3yZGIfzMZnI7KDCx4bkdkL2mi3DSIHrHK6oWo7jJwpRaogC5Nxsw3NJnqOKoav7AyftORtuCEfm2HUui7phr1URfQtlM.4eW-1UESuptiZtQlDq627W3o_9XS9CsuDTufjrhRxdU&dib_tag=se&keywords=car+chassis&qid=1792956400&sprefix=car+chas%2Caps%2C382&sr=8-27

    d. Price – 600 Rs
4. Connector and zero pcb Board and jumper wires
    a. Link - https://www.amazon.in/UNIVERSAL-HUB-General-electronic-Experiment/dp/B0C3B8R4CX?crid=G20W320OA1I4&dib=eyJ2IjoiMSJ9.rvprrkAGoYDmTGGZcpW5-LB_SSX_qKavGwy7gbx-QGJF7kk0xkuVN1XKLsn3NYmE5XCfMSEE4FLf05Ft66lQPR0mwJN1HZngcRh31YkBpQ1sH0ZhXidhxUXohDT28-Ux4XShc341JAhvN7J5jAGsc3LcW7ViuYpdDdtCnIj8b6fFo8jx3hf4BvZofGQPE9tkjL0L3Ai7upn72W0uXyYBwrW4QPvwfK3dXA4_inPigjEg_I1Kftf6PdxaWCGeFMPDJWe9T_i

ycP7fzX8XEMrV8pmcKRtgRaxA3ttXQ4tWEWk.CCd9I4jpdmPiEeVshjx9g_u_gYI5TR9Bh1fYSFlxTaE&dib_tag=se&keywords=pcb&qid=1792956643&sprefix=pcb%2Caps%2C365&sr=8-6

b. **Link -** https://www.amazon.in/UNIVERSAL-HUB-2-54mm-Female-Connector/dp/B0DNMD4THF?crid=1J8VLXGGWHWP&dib=eyJ2IjoiMSJ9.NpiqG_GJTt7YLKhI8CJENxT9AvEhnhfPJXrrYLyOERHSGC4_kxLXNqnx0yyzk-aqy7g2cLjQEYlCnzHF__0pjGKnuop75452prkhQ4znJBnFSVYBcz9eldmtTRidzbQc4DmuNrfiCceRYMJNM6W7eL5pr_Xqyc6JkmTgwn8p1R1btbeRhk5E5Tm8h0gLLnbxJd7xVw9twV3TtPaPuX2JgTNDe5iHCTgdcy52Qe1m_cebNtl-nfkhR3N2oAA6v9CQbXuUWuf6_HwUrBcY5VvgNCYjBU1mlOp7oGow1_iabwk.z5YUZ65MlMrMsfVV9tS6T3RaRvppjT4B6pGOHPkhtww&dib_tag=se&keywords=connector+for+arduino&qid=1792956703&sprefix=connector+for+arduino%2Caps%2C336&sr=8-9

c. **Link -** https://www.amazon.in/Electronic-Spices-Single-Core-Breadboard-Jumper/dp/B0B4WPX449?crid=9X3VZ73B5WV3&dib=eyJ2IjoiMSJ9.vLBL377-brAFNacDz1hApdJzm2dtYtf3ydYQDIMqX9zDlBkcyG_U3TdCXK5vMR9_ddvE36Q8kyzVFZUWfawGi5HX2bVezUOWNIJ5wCM8SDvTGkSNGIQhMOUNQojrLdxHsjpogqVwPZ_fkp7bKSqgTcFXwIpiI6pCDEHFyLO1ZlHowV1UmZL_rZLIJaQ_HRltK4aQQLb7xKn7bq_OYg5ZD2vpYRjQvhvW7r4lnTxXE0EonLFN2yQLBu_zELQ38c8Refl21fFhmgnWo14Cw1M5T8ozV3_z9JL4XUzh_y-4SoI.2GsUc3g-NHeFeNYO9lT6TlBktI4g_z5y3Y-cKrtoV-k&dib_tag=se&keywords=wire&qid=1792956733&sprefix=wire+f%2Caps%2C351&sr=8-10

d. **Link -** https://www.amazon.in/Aptechdeals-Jumper-Wires-Female-Breadboard/dp/B074JB6SX8?crid=U5C7KWHH5QD1&dib=eyJ2IjoiMSJ9.NpiqG_GJTt7YLKhI8CJEN93BeTCKz0zXicFcYBnJD_L-pdUlzupjXnJ_2TqtdUJsGJk-8IcymmMln_8F8uswHeH1QdxGKFiKcD2USV5Tu0hDG8Fz0M_p8cqT91tMxepvU5dn19dCKcgfS0MdT-XWe8XavWfLxDYEtzJEIXGPG3RhQjtzZrpBLPuYUnEfzla6d5Q9cyMx2Wzes7IyJzkFr9NUYjIX5ypTQE4U6F_txL8FZu03rW8NE70vkxjWNH8I8ZmP3-VhhLXpIJgnOdwythShLkztNNB1MVGuvo5PlAI.9e1AUkH8El0hvo9vWiJb-emdn0oyzJONPfReXiNrTzw&dib_tag=se&keywords=jumper+wire+set&qid=1792956577&sprefix=jumper+wire+se%2Caps%2C353&sr=8-6

e. **Price – 60 + 100 + 150 + 140 = 450 Rs**

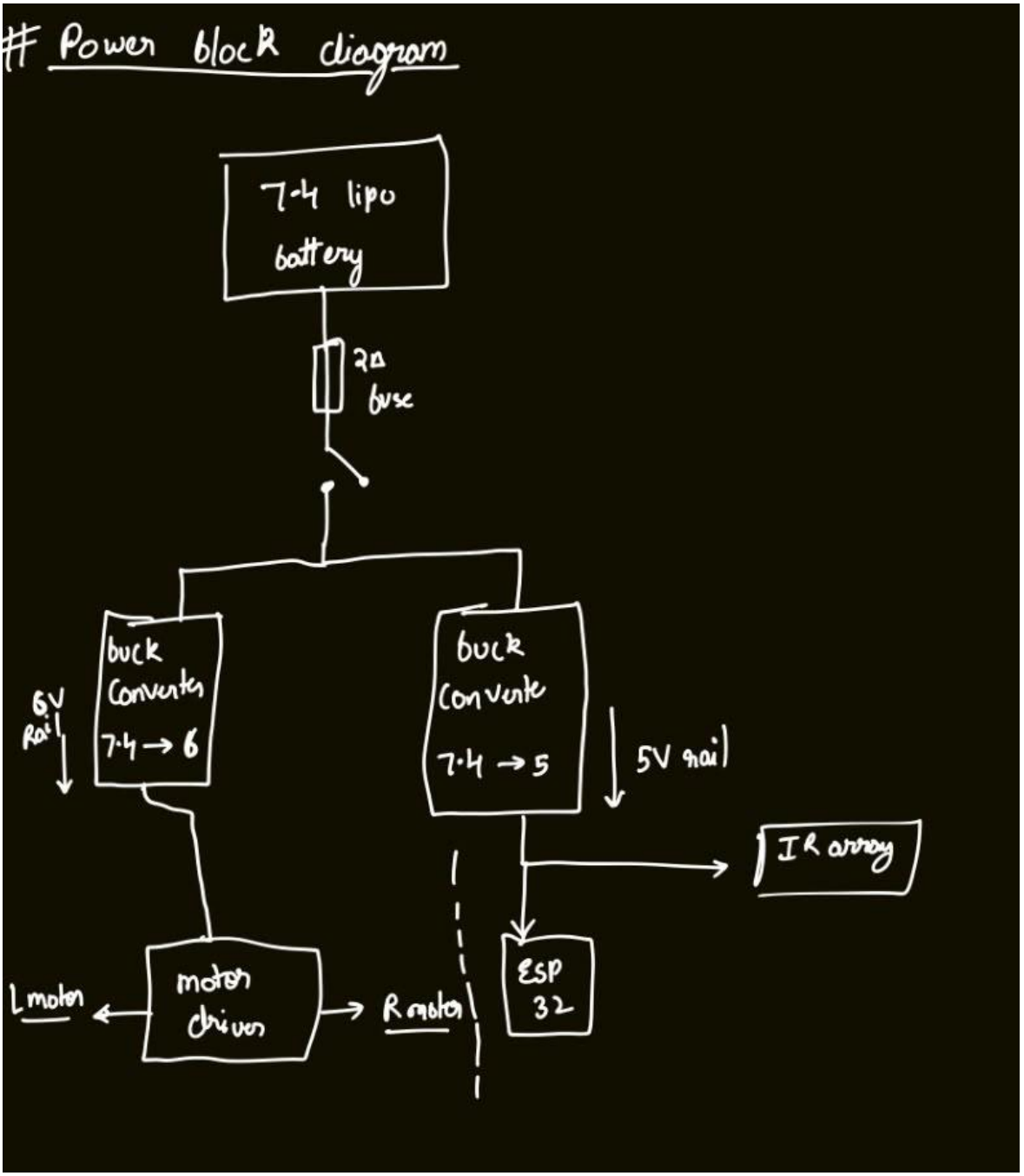
5. **Buck converters ..**
   a. **We need 2 actually one for the 5v and one for the 6v**
   b. **Link -** https://www.amazon.in/Robocraze-Lithium-Battery-Charging-Converter/dp/B07RKH7YVY?crid=1XH81LRF6O6IS&dib=eyJ2IjoiMSJ9.SdP7Hn0UC0hg9sVzZKNKRjp7VQxWtL7Ptks-qowE_IkszxkIlcpUC_XSiOTL10xQI-ea_cagnk7hvw7Dg_b255KHExF-udaxGtXKvYp0t5xakESHLX_v5TA2SA71ODqupyO4HrLPWOgsOUai85we0WdIGsTBUcaXxRRtjEBLO39qWgWPe8J6M7yf4i8BX0SOYf3TtOLdTVKKqxDoPKm8ZK-srQLm8DNyhvAv9tVzTgQ.11TuwI9FQiXiannx1MWrQQTbouhYjeY_Rq356ZCFORE&dib_tag=se&keywords=buck+converters&qid=1793003385&sprefix=buck+converter%2Caps%2C330&sr=8-3
   c. **Price – 2(240) - 480 Rs**

6. **3A Fuse ..**
    a. **Link - https://www.amazon.in/Ovicart-Glass-Fuse-Holder-3A/dp/B09X7B8BM8**
    b. Price – 180 Rs
    c. Reasoning is given where i calculated the max curent draw
7. And ye almost me bhul hi gaya(XT60 connector female and male both)
    a. Link - https://www.amazon.in/Universal-Hub-Pair-XT60-Connector/dp/B0GSBNDQ3L/525-8887240-1156925?pd_rd_w=9DD4Z&content-id=amzn1.sym.d1406b44-aa69-47e4-9270-f613e12d52dc&pf_rd_p=d1406b44-aa69-47e4-9270-f613e12d52dc&pf_rd_r=5NN851TSMB7TNE62N8GB&pd_rd_wg=lUUSA&pd_rd_r=adfcad74-263f-4b85-9296-ea022158de86&pd_rd_i=B0GSBNDQ3L&psc=1
    b. Price – 90 Rs

For Battrey Refer to the Power budget and also the powerblock diagram ..

Task 5 - Power budget and battery selection

Power Block diagram



Battery selection and justification

- We can use the 2S LiPo · 7.4 V · 2200 mAh · 80C battery
    - The main reason why we use lipo batterys is that they deliver high current for their size ..so something like the compact bot of line maze solving ..
    - And also the motors need 6v so its easier to convert 7.4 to 6 and also to 5v basically

- Link - https://www.amazon.in/Original-ABSD-7-4V-2200MAH-Battery/dp/B0GCBRRP1Y?crid=3VTJ8JDSBHGPD&dib=eyJ2IjoiMSJ9.SsgH_-TyyLoZD2CLUlbU0bpgmFQODpvPgVIsCbn-h0va5gWtLmbxCz97PjToHbCnClyJOUIQvBcGzsP3wF2qcVOAWIdjM_Fok_t9YcMLL0V_rLos2MWrSsnzWMsbsHgdEVd-BumfBdXxhihwtLxkucYq3hu3HXNgWjYf-AmXcBezcLUmYJgl0VZ7VcbEvBhqzUsyVmVo_-haakVrdB8Dt3ToiqXtFjcq6AcMqViDUyg.gu161wE9vl6iEiQq6nRAd14dQ8ro1RZQvzNQ2JiEYNQ&dib_tag=se&keywords=2S+LiPo+%C2%B7+7.4+V+%C2%B7+1000+mAh+%C2%B7+25C&nsdOptOutParam=true&qid=1792959097&sprefix=2s+lipo+7.4+v+1000+mah+25c%2Caps%2C370&sr=8-3
- Price – 1600 Rs

Current draw estimation

- Lets talk abt the ratings first then go into calculations
-

Component	Idle	Peak	Basis
ESP32 DevKit	~80 mA	~250 mA	The chip draws 20–68 mA depending on clock speed with the radio off, and 240 mA at Wi-Fi transmit peak. I added ~10 mA for the DevKit's USB-UART chip, LED and regulator (my estimate).
QTR-8RC array	~100 mA	100 mA	LED current is about 20–25 mA per string, giving just under 100 mA for the board. Pololu lists the supply current as 100 mA.
Left N20 motor (6 V)	60 mA	550 mA	One common encoder-equipped N20 lists 0.55 A stall and under 0.06 A no-load at 6 V. SparkFun's N20 pair lists 0.03 A no-load and 0.45 A stall. I used the higher figures.
Right N20 motor (6 V)	60 mA	550 mA	Same as above.
Encoders (both)	~10 mA	~20 mA	Vendors only state a 3.3 V or 5 V supply, not a current. I assumed 5–10 mA each for the Hall sensors and pull-ups.
TB6612FNG	1.5 mA	2.2 mA	Logic current is 1.1–1.5 mA typical and up to 2.2 mA max, and standby is 1 µA max.
Buck converters (×2)	~10 mA	see Step 3	The LM2596 has about 5 mA quiescent current and 80% efficiency at 5 V out, 12 V in, 3 A.

- Summin the currents for rails .. there wll be 2 rails ..5v and 6v rails so

Rail	Idle	Peak
5 V logic (ESP32, QTR, encoders, TB6612 logic)	~192 mA	~372 mA
6 V motors (2 × N20)	120 mA	1,100 mA

- The worst-case total is about 1.5 A (just add the peak currents across all rails)

- But The max current that the battery i selected can give abt 176 A so it might be so that (say in a short more current than the 1.5 A is drawn in fact very high current may be drawn) in that case to protect my comopnets i need a fuse so we need a 3A fuse ....

Runtime calculation

- We are using a 2s 2200 mAh 80c Battery .. so we cannot use 100% of the battery say we use 75% i.e 1650 mAh and coz of the 80c we get 80*2200mAh = 176 A so the battery can supply this much current basically
- Suppose the typical driving and everything works normally and there is not stall and shii so then power of 5v rail is (0.4 + 0.5 + 0.06 ~ 1W) and the 6v rail power is (0.36 + 0.36 ~ 0.72) so total approximately is 1.7W
- So since the battery is 2200 mAh and 7.4 the total energy it has is 2.2* 7.4 but we considering only 75% so 2.2* 7.4(0.75) = 4.7 so we draw approx 1.7 W ideally so approx the car should work for ~ 7h ideally

Cell chemistry justification

-

Property	Li-ion (18650 cells)	LiPo (RC pouch pack)
Energy density	200–265 Wh/kg at cell level	150–200 Wh/kg at pack level
Max discharge rate	2–10C for standard cells	20–120C for RC-grade packs
Cycle life	500–1000 cycles	150–500 cycles
Self-discharge	1–3% per month	2–5% per month
Safety	The metal case makes it more robust to physical damage; LiPo has a medium rating because of puffing.	Soft pouch, so it can swell and be punctured. It usually has no built-in protection.
Shape	Rigid cylinder, needs a holder	Flat and low-profile, easy to fit in a chassis Table of Contents +8

- Why im using a LiPo battery compered to L ion battery
    - It's flat and compact, so it sits low on a small chassis, and you don't need a cell holder.
    - It handles motor start-up spikes and stalls with enormous headroom.
    - Its 7 h typical runtime is more than enough for a robot run.
- The disadvantages
    - **It's oversized.** My worst case is 1.44 A, which is 0.65C on a 2.2 Ah pack, while an 80C pack can deliver up to 176 A. The rating and extra capacity are mostly wasted, and they add weight.
    - **Developing on the big pack and swapping later is going to be my strategy(if the battrey i chose is that bulky then we can change to a smaller one after everything else works).** It gives long test sessions without recharging. It also has low internal resistance